

SUMMATIVE ASSESSMENT - I (SA-1) EXAMINATION (2026-27)

CLASS: VII | SUBJECT: GENERAL SCIENCE

Time Allowed: 2 Hours 30 Minutes | **Maximum Marks:** 80 Marks

General Instructions:

1. *This question paper comprises 5 sections: Section A, Section B, Section C, Section D, and Section E.*
 2. *All questions are compulsory. Internal choices are provided in Section D and Section E.*
 3. *Section A consists of 20 Multiple Choice Questions carrying 1 mark each.*
 4. *Section B consists of 6 Very Short Answer Questions carrying 2 marks each.*
 5. *Section C consists of 7 Short Answer Questions carrying 3 marks each.*
 6. *Section D consists of 3 Long Answer Questions carrying 5 marks each.*
 7. *Section E consists of 3 Source-Based / Case-Based Questions carrying 4 marks each.*
-

SECTION A: OBJECTIVE & MULTIPLE CHOICE QUESTIONS (20 Marks)

Select and write the most appropriate option for each question:

1. Which of the following is an example of a chemical change? [1 Mark]
2. a. Dissolving sugar in water
3. b. Melting of butter in a pan
4. c. Rusting of an iron nail
5. d. Boiling of water into steam
6. When a piece of magnesium ribbon is burned in air, the white ash formed is: [1 Mark]
7. a. Magnesium hydroxide
8. b. Magnesium oxide
9. c. Magnesium carbonate
10. d. Magnesium chloride
11. The three essential conditions required to sustain combustion are: [1 Mark]
12. a. Fuel, Nitrogen, and Heat
13. b. Fuel, Carbon dioxide, and Water
14. c. Fuel, Oxygen, and Heat
15. d. Oxygen, Heat, and Carbon monoxide

16. The protective process of depositing a layer of zinc on iron to prevent rusting is called: [1 Mark]
17. a. Crystallisation
18. b. Galvanisation
19. c. Combustion
20. d. Neutralisation
21. The chemical formula representing rust is: [1 Mark]
22. a. FeO
23. b. Fe_3O_4
24. c. $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
25. d. $\text{Fe}(\text{OH})_2$
26. In a burning candle, which part of the process is considered a physical change? [1 Mark]
27. a. Burning of wax vapor to produce heat and light
28. b. Formation of carbon dioxide gas
29. c. Melting and vaporisation of solid wax
30. d. Formation of black soot
31. Which of the following changes is reversible? [1 Mark]
32. a. Curdling of milk
33. b. Crystallisation of copper sulphate
34. c. Cooking of an egg
35. d. Burning of wood
36. The transitional stage between childhood and adulthood is called: [1 Mark]
37. a. Infancy
38. b. Adolescence
39. c. Senescence
40. d. Toddlerhood
41. The visible protrusion formed by the enlarged larynx in an adolescent boy's throat is called: [1 Mark]
42. a. Thyroid gland
43. b. Epiglottis
44. c. Adam's apple
45. d. Pharynx
46. Which hormone controls the secondary sexual characteristics in boys? [1 Mark]
- a. Insulin
 - b. Oestrogen
 - c. Testosterone
 - d. Thyroxine
47. The master endocrine gland that regulates the hormone production of testes and ovaries is: [1 Mark]
- a. Adrenal gland
 - b. Pituitary gland
 - c. Thyroid gland
 - d. Pancreas
48. The menstrual cycle in human females normally recurs every: [1 Mark]

- a. 7–10 days
 - b. 14–16 days
 - c. 28–30 days
 - d. 45–50 days
49. The first occurrence of menstruation at puberty is termed: [1 Mark]
- a. Menopause
 - b. Menarche
 - c. Ovulation
 - d. Gestation
50. Acne and pimples frequently seen on an adolescent's face are caused by increased activity of: [1 Mark]
- a. Salivary glands
 - b. Sebaceous (oil) glands
 - c. Thyroid glands
 - d. Adrenal glands
51. Complete the analogy: *Testosterone : Boys :: Oestrogen : _* [1 Mark]
- a. Children
 - b. Girls
 - c. Infants
 - d. Elders
52. Which mineral is essential for the formation of hemoglobin in the blood of adolescents? [1 Mark]
- a. Calcium
 - b. Sodium
 - c. Iron
 - d. Iodine
53. Complete the analogy: *Melting of ice : Physical change :: Burning of paper : _* [1 Mark]
- a. Reversible change
 - b. Chemical change
 - c. Periodic change
 - d. Temporary change
54. The national campaign launched by the Government of India against substance abuse is: [1 Mark]
- a. Swachh Bharat Abhiyan
 - b. Nasha Mukta Bharat Abhiyan
 - c. Beti Bachao Beti Padhao
 - d. Digital India
55. **Assertion (A):** Rusting of iron is a chemical change. [1 Mark]
Reason (R): A new substance, hydrated ferric oxide, is formed with completely new properties.
- a. Both (A) and (R) are true and (R) is the correct explanation of (A).
 - b. Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - c. (A) is true, but (R) is false.

- d. Both (A) and (R) are false.
56. **Assertion (A):** Adolescence is characterized by rapid physical growth and hormonal changes. [1 Mark]
Reason (R): Endocrine glands release hormones directly into the bloodstream to trigger maturation.
- a. Both (A) and (R) are true and (R) is the correct explanation of (A).
 - b. Both (A) and (R) are true, but (R) is not the correct explanation of (A).
 - c. (A) is true, but (R) is false.
 - d. Both (A) and (R) are false.
-

SECTION B: VERY SHORT ANSWER QUESTIONS (12 Marks)

Answer each question in one or two sentences:

1. Define a physical change. Give one everyday example. [2 Marks]
 2. State the two necessary environmental conditions required for rusting of iron to occur. [2 Marks]
 3. What is the Fire Triangle? Name its three components. [2 Marks]
 4. Define puberty and mention its typical onset age in human beings. [2 Marks]
 5. What is the difference between menarche and menopause? [2 Marks]
 6. Name the hormone secreted by the thyroid gland and the mineral required for its synthesis. [2 Marks]
-

SECTION C: SHORT ANSWER QUESTIONS (21 Marks)

Answer each question in three to four sentences:

1. Tabulate any three differences between physical changes and chemical changes. [3 Marks]
 2. Explain why curdling of milk is considered an irreversible chemical change. [3 Marks]
 3. Describe the method of galvanisation. Why is it used for iron water pipes and bridge girders? [3 Marks]
 4. Explain how spraying water helps in extinguishing a fire. In what kind of fires should water NOT be used? [3 Marks]
 5. List three secondary sexual characteristics that develop in boys during puberty. [3 Marks]
 6. Why is a balanced diet containing iron, calcium, and proteins essential for adolescents? [3 Marks]
 7. Write a brief note on the pituitary gland and explain why it is called the 'Master Gland'. [3 Marks]
-

SECTION D: LONG ANSWER QUESTIONS (15 Marks)

Answer each question in detail (5 to 7 sentences):

- (a)** Explain the process of burning a candle. Justify how both physical and chemical changes occur simultaneously during this process. [5 Marks]

OR

(b) Describe the chemical reaction that takes place when a magnesium ribbon is burned in air. Write the balanced equation and describe the test to verify the nature (acidic/basic) of the product formed.
- (a)** Describe the biological sequence of events that takes place during the menstrual cycle in human females. What happens if the released egg is not fertilised? [5 Marks]

OR

(b) What are endocrine glands? Name any four endocrine glands found in the human body, the hormone each secretes, and the primary function of each hormone.
- Discuss the physical, mental, and social hazards of substance abuse (drugs and alcohol) among teenagers. Mention the role of the *Nasha Mukh Bharat Abhiyan* in building a healthy youth. [5 Marks]

SECTION E: SOURCE-BASED / CASE-BASED QUESTIONS (12 Marks)

- Case Study 1 (Changes in Kitchen & Garden):** [4 Marks]

Ramesh observed that when an iron pan (*tawa*) was left wet on the kitchen counter for two days, a reddish-brown coating formed on it. In the backyard, he noticed dry leaves turning into rich, dark compost inside an organic pit.

 - a. Name the reddish-brown substance formed on the wet tawa and write its chemical name. [1 Mark]*
 - b. State the two substances from the environment that reacted with the tawa. [1 Mark]*
 - c. Is composting of dry leaves a physical or a chemical change? Justify your answer. [2 Marks]*
- Case Study 2 (Adolescent Growth Spurt):** [4 Marks]

Sneha and Rahul are 13-year-old classmates. Over the past year, Rahul's voice became deep and rough, and a bulge appeared on his neck. Sneha noticed her height increasing rapidly and experienced her first menstrual flow.

 - a. What is the name of the bulge on Rahul's throat, and which organ's enlargement causes it? [1 Mark]*
 - b. Which hormone is responsible for Rahul's deepening voice? [1 Mark]*
 - c. What is the medical term for Sneha's first menstrual period? List two menstrual hygiene guidelines she must follow. [2 Marks]*
- Case Study 3 (Fire Safety & Combustibility):** [4 Marks]

A chemistry teacher demonstrated that paper catches fire when brought near a burner, but wrapping the paper tightly around an iron rod prevents it from burning immediately when held over the same flame.

 - a. What is ignition temperature? [1 Mark]*

- *b. Why does paper wrapped around an iron rod not catch fire quickly? [2 Marks]*
- *c. Which component of the Fire Triangle is affected in this demonstration? [1 Mark]*